

SignWriting OneD — regeneration report

Comparison of the upstream Sutton SignWriting OneD font vs the font regenerated from font-db's cubic-Bezier SVG sources, with and without our ellipse-replacement optimization.

File size

variant	size		Δ
original (upstream)	8,206,192 bytes	8013.9 KB	
new (no optimisation)	7,537,524 bytes	7360.9 KB	-8.1% vs original
new (ellipse + rotation dedup)	2,668,892 bytes	2606.3 KB	-67.5% vs original

Glyph count

variant	glyphs
original (upstream OneD)	38,316 mapped codepoints
new (no optimisations)	38,316 mapped codepoints
new (ellipse + rotation dedup)	38,316 mapped codepoints

Every SignWriting symbol ``S{base}+rot{rot}` shares an outline with 7 other drawings — the last hex digit picks one of 16 orientations (8 rotations of rot 0, plus their mirrors at the rot-8 offset). Diagonals (odd indices) form a parallel sub-family centred on rot 1.

Rotation + reflection encoding

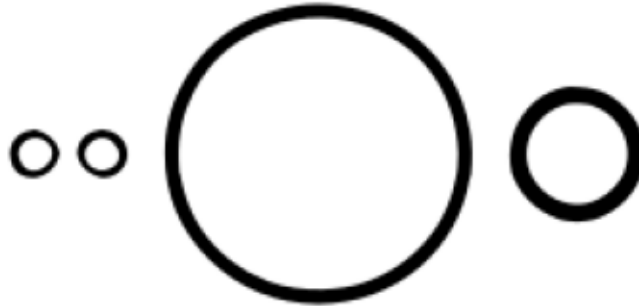
last hex	semantic orientation	composite source	transform applied
0	rot 0°	base (outline kept)	—
1	rot 45°	diag base (outline kept)	—
2	rot 90°	← rot 0	rotate 90°
3	rot 135°	← rot 1	rotate 90°
4	rot 180°	← rot 0	rotate 180°
5	rot 225°	← rot 1	rotate 180°
6	rot 270°	← rot 0	rotate 270°
7	rot 315°	← rot 1	rotate 270°
8	mirror + rot 0°	← rot 0	mirror
9	mirror + rot 45°	← rot 1	mirror
A	mirror + rot 90°	← rot 0	mirror + rot 270°
B	mirror + rot 135°	← rot 1	mirror + rot 270°
C	mirror + rot 180°	← rot 0	mirror + rot 180°
D	mirror + rot 225°	← rot 1	mirror + rot 180°
E	mirror + rot 270°	← rot 0	mirror + rot 90°
F	mirror + rot 315°	← rot 1	mirror + rot 90°

Mirror reverses CCW→CW, which is why rot A's matrix is ``mirror + rot 270°`` even though the semantic is ``mirror + rot 90°``.

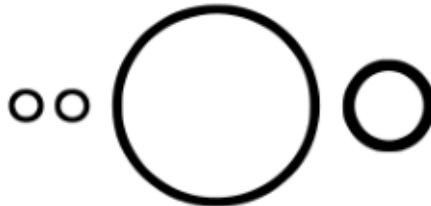
Ellipse optimization — circular glyphs

hand-traced cubic circles replaced by 4-segment Bezier ellipses

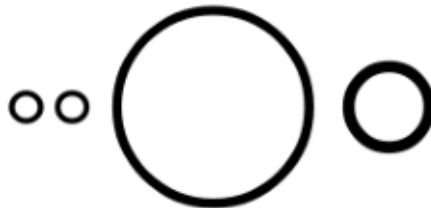
original (upstream OneD)



new (no optimisations)



new (ellipse + rotation dedup)



Rotation dedup — all 16 orientations of S1000{0..f}

even indices = composites of rot 0; odd = composites of rot 1

original (upstream OneD)



new (no optimisations)



new (ellipse + rotation dedup)

